Reference Material

DS-RX1 Print System Flow

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Dai Nippon Printing Co., Ltd.

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1. Configuration

(1) Single Printer System

Printers : One unit
Buffering : Double Buffer
Spooling : Spool ON



[Sample Flow]

The basic flow of the print process is to check the printer's status before sending data, and to send the data when the printer is ready. The sequence is as follows:

(1) Upon receipt of a print request, checks if there is already a print job (WindowsAPI EnumJobs), and if there is already a print job found, loops to check the status of the print job.
(If there is already a print job when Status API tries to get the printer status, the application is obliged to wait until the print job is completed and the printer status can be obtained. In this case, it's more effective to loop to check the number of print jobs until the print job is cleared.)

(2) Checks for errors, and if there is any error found, goes to error recovery process (Refer to 2 : Error Recovery Flow).

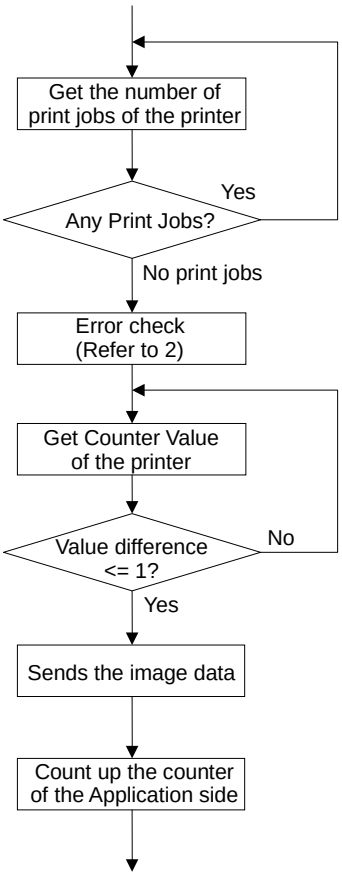
(3) If there is no print job nor any error, gets counter value of the printer (Status API GetCounterL). Then checks if there are any free buffers in the printer by comparing the counter values of the application side and the printer side.

Counter value difference = 0 --> 2 free buffers
Counter value difference = 1(*1) --> 1 free buffer
Counter value difference = 2(*2) --> 0 free buffers

(4) When there is a free buffer, sends image data to the printer.

(5) After sending out the image data, counts up the counter value of the application side.

(6) If there are no free buffers, loops to check status for free buffers.

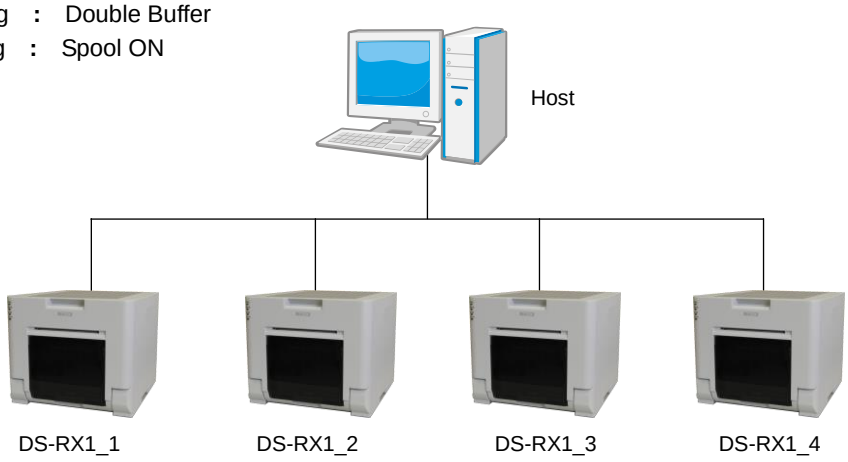


(*1) When printing 6x8(A5), 5x7(2L), 6x6(*4), 5x5(*4) or the dual image of 6x4 or 5x3.5(*3) size, the counter value difference will be 2.
(*2) When printing 6x8(A5), 5x7(2L), 6x6(*4), 5x5(*4) or the dual image of 6x4 or 5x3.5(*3) size, the counter value difference will be 4.

*3. Effective at firmware Ver 1.20 or later.
*4. Effective at firmware Ver 2.06 or later.

(2) 4 Printer System

Printers : Four units
Buffering : Double Buffer
Spooling : Spool ON



[Sample Flow]

The basic flow of the print process is to check for a printer that is ready before sending data, and to send the data to the available printer. The sequence is as follows:

(1) Upon receipt of a print request, checks if the current printer has a print job (WindowsAPI EnumJobs), and if there is a print job, checks for another printer's availability.
(If there is already a print job when Status API tries to get the printer status, the application is obliged to wait until the print job is completed and the printer status can be obtained. In this case, when there is a print job found in the current printer, it's more effective to search for another printer that is ready.)

(2) Checks for errors, and if there is any error found, goes to error recovery process (Refer to 2 : Error Recovery Flow).

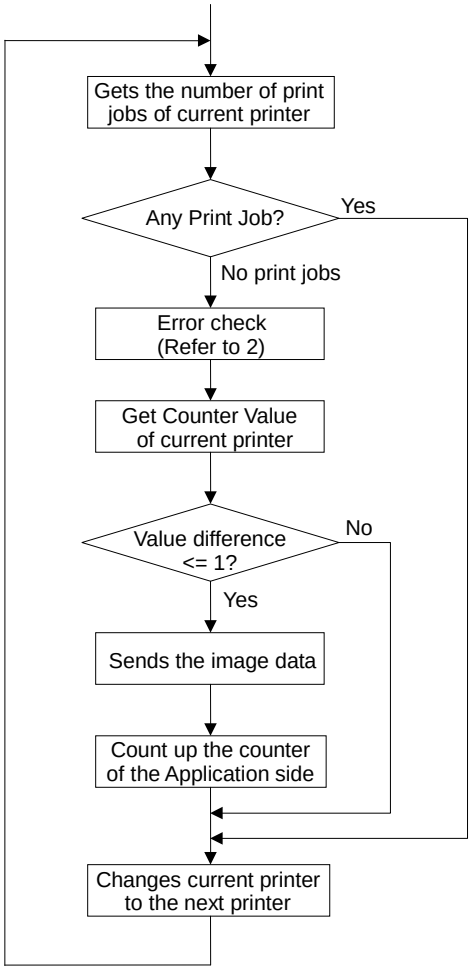
(3) If there is no print job nor any error, gets counter value of the printer (Status API GetCounterL).
Then checks if there are any free buffers in the printer by comparing the counter values of the application side and the printer side.

Counter value difference = 0 --> 2 free buffers
Counter value difference = 1(*1) --> 1 free buffer
Counter value difference = 2(*2) --> 0 free buffers

(4) When there is a free buffer, sends image data to the printer.

(5) After sending out the image data, counts up the counter value of the application side, and changes the current printer.

(6) If there are no free buffers, changes the current printer and searches another printer that is ready.



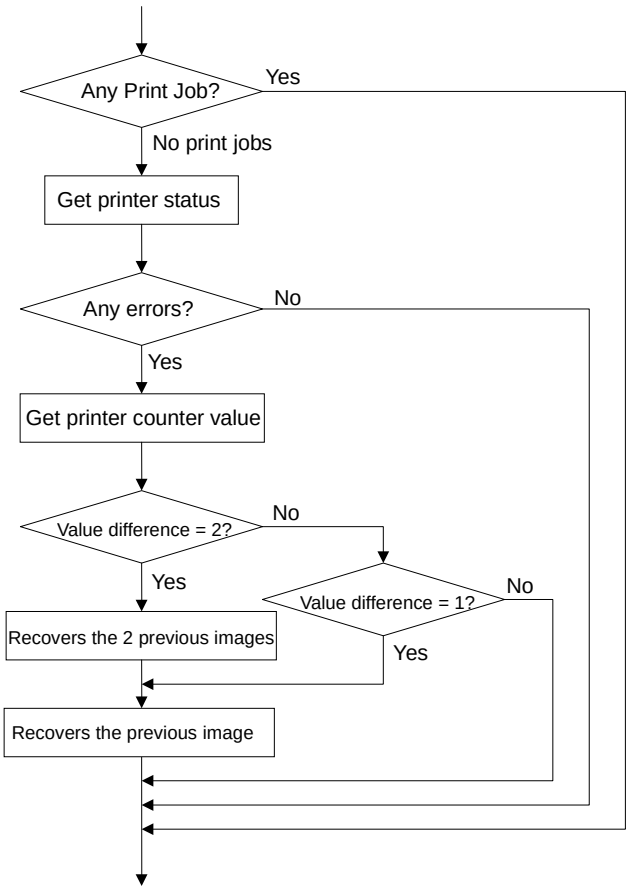
2. Error Recovery Flow

When an error occurs, it's necessary to find the image that was being printed at that time to find out the right image(s) to be recovered, based on the printer's counter.

(1) Checks for print jobs, and if there are any print jobs, does not get the status of the printer. (*3)
(2) If there are no print jobs, gets the status of the printer and checks for any errors.
(3) If there is any error found, gets counter value of the printer.
(4) Compares the value of the printer's counter and the counter of the application side.
(5) If the difference of the two counter values is 2(*2), recovers the two previous images. (In a multi-printer system, sends the image to be reprinted to another printer that is ready. If it is a single printer, sends the image to be reprinted after the printer error is solved.) In the case when the difference is 1(*1), recovers the previous image.

[Note]
- In the case where a print job is stacked or is not released for a long time, it's possible there is a communication error, and it may be necessary to go into the error recovery process.
- If there is no print job and communication is suspended for some reason, a failure code will be returned approx. 1 minute after sending the DS-RX1 printer status API to get the printer status. (During this time, the application is idling, waiting for the status to be returned) So, when confirming the status, it's recommended to check the time from the start of the status check to the end. If the time in between is too long, it's necessary to go into the error recovery process.

(*3) In a situation where there is a print job, the printer spooler has priority to access the USB port.
During this time, if the printer status is checked by DS-RX1 printer status API, API will be stopped until the print job is completed, since the USB port cannot be accessed.
So, before checking printer status by DS-RX1 printer status API, please check whether there are any print jobs left or not first, and check the printer status only when there is no print job.



3. Concerning Error Onset and Life-counter Count Up Timing

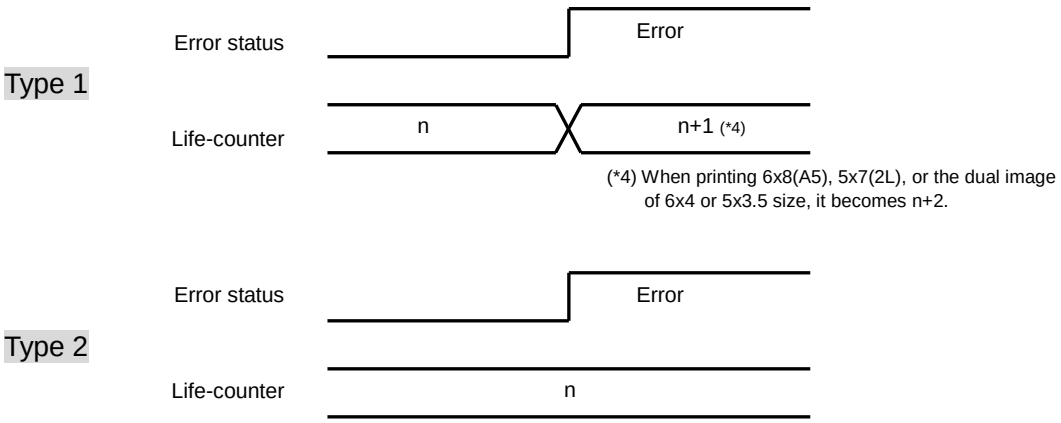
When an error occurs, the timing of the count up for the life-counter is as shown here.
There are two kinds of timing, as shown in [Type 1] and [Type 2] below.

■Type 1

Ribbon end and paper end, and the error status is set after completion of printing. Because the printing has been completed normally, the life-counter value is counted up at the same time that the error status is set.

■Type 2

Any error other than ribbon end and paper end. Because printing is not complete at the time that the error status is set, the life-counter value is not counted up. (This is errors like ribbon cut and cutter error.)



4. Error Recovery

There are two kinds of errors; errors detected when a printer error status is obtained, and errors where a print job is kept waiting in the queue.
Normally, most errors are detected when a printer error status is obtained. Errors where a print job is kept waiting in the queue occur when there is a communication problem, as when the USB cable is disconnected or the printer power turned OFF during transmission.

[Error Status errors]

- When an error is detected, the printer's life-counter value is obtained.
- The count value from the application and the printer's life-counter value are compared.
- Using the difference in count values, the data that needs to be reprocessed is determined.
  [difference is 0] : Printing of all transmitted data was completed, so data doesn't need to be sent again.
  [difference is 1(*2)]: Printing of the previous 1 data was not completed, so that data needs to be reprocessed.
  [difference is 2(*3)]: Printing of the previous 2 data was not completed, so both those data need to be reprocessed.

[Print Job errors]

- For errors where a print job is kept waiting in the queue, the error type and the life-counter value cannot be obtained through transmission. Therefore, when an error occurs, a message will appear indicating a problem with the printer. The reason for the error must be checked manually, and resolution of the problem must also be performed manually.
- When the error has been solved, the printer's life-counter will be obtained and compared to the count value from the application.
- Using the difference in count values, the data that needs to be reprocessed is determined. (See above)